Q2

工业环境电缆的选择与截面确定
Cable selection and sizing in an industrial environment

工业电缆应同时适合负载、安装方式和外部环境。选择内容包括导体材料、芯数、绝缘、铠装及护套；截面则须同时满足载流量、压降、故障保护及损耗要求。额定电流相同的回路，因敷设和环境不同，所需电缆也可能不同。

Industrial cables must suit the load, installation arrangement and external environment. Selection covers conductor material, number of cores, insulation, armour and sheath. The conductor size must satisfy current-carrying capacity, voltage drop, fault protection and loss requirements together. Circuits with the same rated current may require different cables under different conditions.

1 三组表的应用 / Applying the three table groups

题目指定的表 / Table group	作用	Function
Cable correction factors 电缆修正系数	根据实际环境温度、成组及隔热条件修正表列载流量，并考虑保护器件所需的修正。	Adjust the tabulated capacity for actual ambient temperature, grouping and thermal insulation, together with the correction required by the protective-device type.
Table A6(1) 电缆载流量及压降表	按电缆类型、截面、载流导体及对应安装栏位查载流量，并读取相应压降数据。	Provides current-carrying capacities and voltage-drop data for the stated cable type, cross-section, conductor arrangement and installation column.
Appendix 7 电缆安装方式	将现场实际安装情况对应到参考安装方法，使查表条件与实际布置相符。	Matches the actual arrangement to a reference installation method so that the rating column corresponds to the installation.

Table A6(1) 的适用条件 / Scope of Table A6(1)

Table A6(1)适用于铜导体、单芯、PVC 绝缘、非铠装电缆，可带或不带护套；表列参考环境温度为 30°C，导体运行温度为 70°C。查表时还须匹配单相或三相、载流导体数量及安装条件。

Table A6(1) covers single-core, PVC-insulated, non-armoured copper cables, with or without sheath, at a reference ambient temperature of 30°C and conductor operating temperature of 70°C. The phase arrangement, number of loaded conductors and installation conditions must also match.

采用铝导体、多芯、XLPE 绝缘或铠装结构时，应选择相应类别的额定资料；再结合适用的修正系数及压降数据确定截面。

For aluminium, multicore, XLPE-insulated or armoured constructions, use the corresponding rating data, then apply the relevant corrections and voltage-drop checks to determine the size.

2 负载及电缆结构 / Loading and cable construction

因素 / Factor	定性解释	Qualitative explanation
1 Design current (Ib) 设计电流	应按实际机械设备、最大需量、同时使用情况和工作周期确定设计电流，并考虑功率因数及效率。电动机启动还须检查短时电流和启动压降，保护特性应适合启动过程。	Determine Ib from actual machinery loads, maximum demand, simultaneous operation and duty cycle, allowing for power factor and efficiency. For motors, also check short-duration starting current and starting voltage drop, and select protection suited to starting.
2 Conductor material 导体材料	铜和铝的电阻率、机械特性及连接要求不同。同等负载和安装条件下，铝导体通常需要较大的截面积；还须兼顾重量、成本及接线端子的适配。	Copper and aluminium have different resistivity, mechanical properties and connection requirements. Aluminium generally requires a larger cross-sectional area for comparable loading and installation conditions. Weight, cost and terminal compatibility also affect the choice.
3 Insulation type and operating temperature 绝缘类型及运行温度	PVC 与 XLPE 的允许导体温度不同，应按选用的绝缘类型查表。较高的绝缘温度等级还须与接线端子及设备的温度限值协调，不能仅凭绝缘类型提高整条回路的载流量。	Select the rating table for the actual insulation, such as PVC or XLPE. Its permissible conductor temperature must also be compatible with terminals and connected equipment; a higher insulation temperature rating alone does not establish a higher rating for the complete circuit.
4 Number of cores and circuit arrangement 芯数及回路排列	单芯或多芯、单相或三相、实际载流导体数量及排列都会影响发热和查表栏位。非线性工业负载还可能使中性导体承受明显谐波电流，其发热和截面应单独评估。	Single-core or multicore construction, phase arrangement, the number of loaded conductors and their disposition affect heating and the applicable table column. Nonlinear industrial loads may also produce significant neutral harmonic current, requiring assessment of neutral heating and size.
5 Armour and sheath 铠装及护套	应按机械防护、潮湿及腐蚀等条件选择铠装和护套，并使用对应电缆结构的额定资料。交流单芯电缆的铠装及金属附件还须考虑磁性发热。	Choose armour and sheath for the mechanical, moisture and corrosive conditions, using rating data for that cable construction. For single-core a.c. cables, the armour and associated metalwork must also be suitable for the magnetic heating effects of that arrangement.
6 Voltage rating and supply system 额定电压及供电系统	电缆的额定电压应适合系统电压和对地电压。交流或直流、单相或三相及负载功率因数，亦决定设计电流、导体配置和压降数据的选择。	The cable voltage rating must suit the system voltage and voltage to earth. A.c. or d.c. operation, phase arrangement and load power factor also govern design current, conductor configuration and selection of voltage-drop data.

3 安装条件及载流量 / Installation and current-carrying capacity

因素 / Factor	定性解释	Qualitative explanation
7 Installation methods 安装方式	应按 Appendix 7 识别实际安装情况，再选对应参考方法。穿管、线槽、直接固定、桥架或自由空气的散热条件不同。同一线路须检查限制载流量的区段；封闭地沟或地板混凝土槽还可能需要专用修正。	Identify the actual arrangement in Appendix 7 and use the corresponding reference method. Conduit, trunking, clipped-direct, tray and free-air installations dissipate heat differently. Assess the limiting route section; enclosed trenches or infloor concrete troughs may also require specific rating factors.
8 Ambient temperature (Ca) 环境温度修正系数	应采用路线中实际可能出现的环境温度，例如炉旁或高温厂房区域。环境越热，导体可利用的温升余量越小，Ca 通常越低，因而可能需要增大截面或改善路线。	Use the ambient temperature expected along the route, including hot production areas or locations near furnaces. Higher ambient temperature reduces the available temperature-rise margin and normally lowers Ca, potentially requiring a larger conductor or a better route.
9 Grouping (Cg) 成组修正系数	多条同时带载回路在管槽内或成束敷设，会互相加热并降低每条电缆的载流量。应按回路数量、排列和间距选择 Cg；分开敷设或增加间距可改善散热。	Loaded circuits grouped in an enclosure or bunched together heat one another and reduce each cable's capacity. Select Cg for the number of circuits, arrangement and spacing. Separating circuits or increasing spacing can improve heat dissipation.
10 Thermal insulation (Ci) 隔热修正系数	电缆被建筑隔热材料包围时，热量较难散出。影响取决于包围程度、长度及散热接触条件。应采用适用的 Ci，并确认参考方法已包含哪些热条件，避免重复折减。	Building thermal insulation restricts heat loss from a cable. The effect depends on the extent and length of enclosure and the available heat-transfer path. Apply the appropriate Ci, checking which thermal effects are already included in the reference method to avoid double counting.
11 Protective-device rating (In) and Cd 保护器件额定值及修正	电缆应与过载保护额定值或整定值协调。Cd 在采用 BS 3036 半封闭熔断器的课程选线方法中为 0.725，其他所列类型通常为 1。电动机回路须适当配合启动与过载保护。	Coordinate the cable with the overload protective-device rating or setting. In the course sizing method, Cd is 0.725 for a BS 3036 semi-enclosed fuse and normally 1 for the other listed types. Motor starting and overload protection must be coordinated appropriately.

修正系数与保护配合 / Corrections and protection coordination

$$I_b \le I_n \le I_z \qquad I_t \ge I_n \div (C_a \times C_g \times C_i \times C_d)$$

Ib 为设计电流；In 为过载保护额定值或整定值；Iz 为安装后的载流量；It 为表列载流量。仅采用实际适用的修正。

Ib: design current; In: overload-protection rating or setting; Iz: installed capacity; It: tabulated capacity. Apply only the relevant correction factors.

4 电气性能及保护 / Electrical performance and protection

因素 / Factor	定性解释	Qualitative explanation
12 Cable length and voltage drop 电缆长度及电压降	线路越长、负载电流越大，压降通常越大；导体电阻、电抗及功率因数亦有影响。长距离回路即使载流量合格，也可能须增大截面，以保证设备端电压及电动机启动性能。	Longer routes and higher load current generally increase voltage drop; conductor resistance, reactance and power factor also matter. A long circuit may require a larger conductor even when its thermal rating is adequate, to maintain terminal voltage and motor-starting performance.
13 Short-circuit withstand 短路承受能力	导体和绝缘必须承受故障电流在切除前产生的热应力。预期故障电流越大、切除时间越长，要求越严格；保护器件的分断能力亦须足以切断该故障电流。	The conductor and insulation must withstand the thermal stress produced before a fault is cleared. Higher prospective fault current and longer clearing time increase the duty. The protective device must also have sufficient breaking capacity to interrupt that fault current.
14 Earth-fault protection and protective conductor 接地故障保护及保护导体	应核查故障环路阻抗、保护导体连续性及所需切除条件。线路过长或导体过小会增加环路阻抗；相导体和保护导体的截面须与故障保护配合。	Check earth-fault loop impedance, protective-conductor continuity and the required disconnection conditions. Long routes or small conductors increase loop impedance. Phase and protective-conductor sizes must be coordinated with the fault-protection arrangement.
15 Power loss and energy efficiency 功率损耗及能源效益	工业设备可能长时间带载运行，导体电阻引起的 I^2R 损耗会形成持续电费及发热。较大截面通常可降低电阻和损耗，应综合初期成本、使用时间及运行成本选择。	Industrial equipment may operate under load for long periods. Conductor I^2R loss causes continuing energy cost and heating. A larger cross-sectional area generally reduces resistance and losses, so initial cost should be assessed together with operating hours and energy cost.

压降应合并检查 / Check the complete voltage-drop budget

应把上游与下游压降合并检查。在本课程采用的 4%总压降条件下，如果上游已使用 1%，下游可用余量便为 3%。表列 mV/A/m 值须与相制、导体排列及安装条件匹配。电动机回路还须检查启动期间的电压下降。

Assess upstream and downstream voltage drop together. Under the 4% total limit used in the course, an upstream drop of 1% leaves 3% for the downstream circuit. Select the mV/A/m data for the correct phase arrangement, conductor disposition and installation. Motor circuits also require a starting-voltage check.

因此，截面应满足所有适用条件：载流量合格后，若压降、故障承受能力或保护配合仍不满足，便须增大截面、改善路线或调整设计。

The size must therefore satisfy every applicable condition. After meeting the thermal rating, an unsatisfactory voltage drop, fault withstand or protection arrangement still requires a larger conductor, an improved route or another design adjustment.

5 工业环境适用性 / Suitability for industrial conditions

因素 / Factor	定性解释	Qualitative explanation
16 External influences 外部环境影响	潮湿、水、油、化学品、腐蚀性气体、粉尘及外部热源会影响绝缘和护套寿命。应选择相容材料、密封及防护方式；增加截面并不能代替所需的环境适用性。	Moisture, water, oils, chemicals, corrosive atmospheres, dust and external heat can damage insulation or sheath. Select compatible materials, sealing and protection. A larger conductor cross-section does not replace the need for environmental suitability.
17 Mechanical stresses 机械应力	冲击、磨损、振动、拉力及反复移动会影响电缆结构与支承要求。固定厂房线路可采用合适铠装或管槽保护；移动机械须采用适合反复弯曲的电缆及连接方式。	Impact, abrasion, vibration, tension and repeated movement govern cable construction and support. Fixed factory circuits may need appropriate armour or containment. Moving machinery requires cables and connections suitable for repeated flexing.
18 Fire and explosion conditions 火灾及爆炸条件	应考虑火焰蔓延及烟气影响。消防或应急回路还须考虑规定条件下的供电连续性。存在可燃气体或粉尘时，电缆、进线装置及布线系统须适合相应危险条件。	Consider flame spread and smoke effects. Fire-service or emergency circuits may also require continuity of supply under specified fire conditions. Where flammable gases or dusts are present, cables, entries and the wiring system must suit the hazardous conditions.
19 Terminations and installation feasibility 端接及安装可行性	截面及电缆外径应与端子、接线耳、格兰头和管槽容量匹配，并满足弯曲、牵引和支承条件。端接不良会产生局部过热；路线也应便于检查、测试和维修。	The conductor size and cable diameter must suit terminals, lugs, glands and containment capacity, as well as bending, pulling and support conditions. Poor terminations cause local overheating. The route should also permit inspection, testing and maintenance.
20 Future loading and alterations 未来负荷及改造	如有明确扩建计划，应预留合理的负荷及布线空间。新增回路或提高保护整定值后，必须重新检查负载、成组修正、压降及故障条件，以保持整条回路协调。	Where expansion is planned, allow reasonably for future load and cable space. Adding circuits or increasing protection settings requires reassessment of loading, grouping, voltage drop and fault conditions so that the complete circuit remains coordinated.

选择顺序 / Selection sequence

先确定实际负载及设计电流，再选择适合工业环境的电缆结构和安装方法；随后按正确表格及适用修正系数核查载流量，并检查压降、故障保护、损耗及端接条件。最终选用同时满足这些要求的规格。

Establish the actual load and design current, then select the cable construction and installation method for the industrial environment. Check current-carrying capacity using the correct table and applicable corrections, followed by voltage drop, fault protection, losses and termination conditions. Select a size that satisfies these requirements together.